Sricharan Balaji 240801328 week 7

Question **1**
Correct
Marked out of 5.00
Flag question

Sunny and Johnny like to pool their money and go to the ice cream parlor. Johnny never buys the same flavor that Sunny does. The only other rule they have is that they spend all of their money.

Given a list of prices for the flavors of ice cream, select the two that will cost all of the money they have.

For example, they have $m = 6$ to spend and there are flavors costing $cost = [1, 2, 3, 4, 5, 6]$. The two flavors costing 1 and 5 meet the criteria. Using 1-based indexing, they are at indices 1 and 4.

Function Description

Complete the code in the editor below. It should return an array containing the indices of the prices of the two flavors they buy.

It has the following:

- m: an integer denoting the amount of money they have to spend
- cost: an integer array denoting the cost of each flavor of ice cream

Input Format

The first line contains an integer, t, denoting the number of trips to the ice cream parlor. The next t sets of lines each describe a visit. Each trip is described as follows:

1. The integer m, the amount of money they have pooled.
2. The integer n, the number of flavors offered at the time.
3. n space-separated integers denoting the cost of each flavor: $cost[cost[1], cost[2], . . . , cost[n]]$.

Note: The index within the cost array represents the flavor of the ice cream purchased.

Constraints

- $1 \le t \le 50$
- $2 \le m \le 10^4$
- $2 \le n \le 10^4$
- $1 \le cost[i] \le 10^4$, " i Î [1, n]
- There will always be a unique solution.

Output Format

For each test case, print two space-separated integers denoting the indices of the two flavors purchased, in ascending order.

Sample Input

```
2
4
5
1 4 5 3 2
4
4
2 2 4 3
```

Sample Output

```
1 4
1 2
```

Explanation

Sunny and Johnny make the following two trips to the parlor:

1. The first time, they pool together $m = 4$ dollars. Of the five flavors available that day, flavors 1 and 4 have a total cost of $1 + 3 = 4$.
2. The second time, they pool together $m = 4$ dollars. TOf the four flavors available that day, flavors 1 and 2 have a total cost of $2 + 2 = 4$.

```c
#include <stdio.h>
int main(){
	int t,n,m;
	scanf("%d",&t);
	while(t--){
		scanf("%d",&m);
		scanf("%d",&n);
		int arr[n];
		for(int i=0;i<n;i++)scanf("%d",&arr[i]);
		for(int i=0;i<n;i++){
			for(int j=i+1; j<n;j++){
				if (arr[i]+arr[j]==m)printf("%d %d\n",i+1,j+1);
			}
		}
	}
	return 0;
}
```

	Input	Expected	Got	
✓	2 4 5 1 4 5 3 2 4 4 2 2 4 3	1 4 1 2	1 4 1 2	✓

Passed all tests! ✓

Question **3**
Correct
Marked out of 5.00
Flag question

Watson gives Sherlock an array of integers. His challenge is to find an element of the array such that the sum of all elements to the left is equal to the sum of all elements to the right. For instance, given the array ***arr = [5, 6, 8, 11]***, ***8*** is between two subarrays that sum to ***11***. If your starting array is ***[1]***, that element satisfies the rule as left and right sum to ***0***.

You will be given arrays of integers and must determine whether there is an element that meets the criterion.

Complete the code in the editor below. It should return a string, either YES if there is an element meeting the criterion or NO otherwise.

It has the following:

- arr: an array of integers

Input Format

The first line contains ***T***, the number of test cases.

The next ***T*** pairs of lines each represent a test case.

- The first line contains ***n***, the number of elements in the array ***arr***.

- The second line contains ***n*** space-separated integers ***arr[i]*** where $0 \le i < n$.

Constraints

- $1 \le T \le 10$
- $1 \le n \le 10^5$
- $1 \le arr[i] \le 2 \times 10^4$
- $0 \le i \le n$

Output Format

For each test case print YES if there exists an element in the array, such that the sum of the elements on its left is equal to the sum of the elements on its right; otherwise print NO.

Sample Input 0

```
2
3
1 2 3
4
1 2 3 3
```

Sample Output 0

```
NO
YES
```

```c
#include <stdio.h>
int main()
{
    int t,n,lsum=0,rsum=0,k=0;;
    scanf("%d",&t);
    while(t--){
        scanf("%d",&n);
        int arr[n];
        for(int i=0;i<n;i++){
            scanf("%d",&arr[i]);
            lsum+=arr[i];}
        for(int j=n-1;j>=0;j--){
            lsum-=arr[j];
            if(lsum==rsum) k++;
            rsum+=arr[j];
        }
        if(k!=0) printf("YES\n");
        else printf("NO\n");
    }
    return 0;
}
```

	Input	Expected	Got	
✓	3 5 1 1 4 1 1 4 2 0 0 0 4 0 0 2 0	YES YES YES	YES YES YES	✓
✓	2 3 1 2 3 4 1 2 3 3	NO YES	NO YES	✓

Passed all tests! ✓